Muhammad Waqar

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Education

University of Engineering and Technology (UET) Oct. 2015 - Aug. 2019
Bachelor of Science in Computer Engineering *Taxila, Punjab*

Experience

Softoo May 2025 - Present
AI Lead Engineer: Generative AI | RAG Systems | AI Agents *Islamabad, Punjab*

- Leading AI agentic systems development and RAG (Retrieval-Augmented Generation) pipeline implementation for intelligent enterprise automation and decision-making platforms.
- Architecting multi-agent AI frameworks with autonomous reasoning capabilities, enabling complex task orchestration and self-improving system behaviors.
- Designing and deploying advanced RAG architectures with vector databases, semantic search, and context-aware knowledge retrieval for enterprise knowledge management systems.
- Specialized in local deployment of large language models including SQLCoder, Llama, and Code Llama for secure on-premise AI solutions, ensuring data privacy and reduced inference costs.
- Developed comprehensive AI-powered audit systems covering BCM, SAMA, and ISO compliance domains for automated compliance checking, risk assessment, and regulatory validation, streamlining audit processes through intelligent document analysis and anomaly detection.
- Mentoring junior engineers and establishing best practices for AI agentic development lifecycle and model governance in production environments.

Forbmax (pvt) Ltd May 2023 - May 2025
AI Team Lead: Computer Vision | NLP | Data Science *Islamabad, Punjab*

- Led a team of machine learning professionals, providing strategic guidance and technical mentorship to drive innovative solutions and collaborative efforts across multiple AI domains.
- Designed and developed advanced machine learning models for mission-critical applications, ensuring high performance and reliability with comprehensive testing and validation frameworks.
- Created and optimized end-to-end model pipelines, streamlining data preprocessing and model deployment processes, improving operational efficiency by 40% through automated workflows.
- Implemented custom training frameworks and fine-tuning strategies for transformer models, YOLO architectures, and computer vision models to achieve domain-specific performance improvements.
- Spearheaded fine-tuning of large language models and vision transformers using transfer learning techniques, achieving significant accuracy improvements on specialized datasets and business-specific use cases.
- Automated the annotation process through development of intelligent labeling systems and active learning frameworks, reducing manual annotation effort by 70% while maintaining data quality and consistency.
- Championed the design of robust APIs and microservices architecture, enhancing product accessibility and ensuring seamless integration with external systems and platforms for scalable AI solutions.
- Managed the successful deployment of machine learning solutions in production environments, optimizing real-time processing capabilities and reducing operational costs through efficient resource utilization.

Horizon Tech Services Jun. 2020 - April 2023
Sr.AI Engineer *Islamabad, Punjab*

- Served as a Senior AI Engineer, responsible for leading end-to-end machine learning projects while providing strategic guidance and mentorship to a dedicated team of professionals.
- Managed and supervised data creation, preprocessing, model training, and deployment for a variety of applications, ensuring data quality and model effectiveness.
- Oversaw extensive data handling, including data collection, cleaning, and annotation, to support the development of high-performing machine learning models.
- Designed and implemented robust data preprocessing pipelines, optimizing feature extraction and data augmentation techniques for enhanced model performance.
- Led the fine-tuning and optimization of models, with a particular focus on object detection, resulting in improved object recognition and tracking.

- Managed the deployment of machine learning solutions in production environments, collaborating cross-functionally to guarantee seamless integration and real-time processing.

Lampro Mellon IT Solutions Pvt. Ltd Aug. 2019 - May 2020

Python Developer *Lahore, Punjab*

- Worked as a Junior Python Developer, contributing to the development of software components with a focus on verification and high-level programming design.
- Collaborated with experienced team members to design and implement software modules, ensuring they meet quality and reliability standards.
- Assisted in the creation of automated verification scripts and tools to streamline the testing and validation processes.
- Participated in code reviews to learn and improve software quality and coding practices.
- Contributed to the design and architecture of software systems, applying theoretical knowledge gained during education.
- Worked on algorithm implementation and data processing, applying problem-solving skills to real-world challenges.

Projects

Enterprise Compliance AI Agent Platform | *Agentic AI, RAG, LangGraph* Sep 2026

- Designed and implemented an agentic AI audit platform covering BCM, SAMA, and ISO compliance domains, including automated maturity assessments across policies, contracts, and internal documentation.
- Built an end-to-end audit workflow that plans engagements, generates risk and control matrices, executes control tests, and produces audit-ready reports, orchestrated with LangGraph and LangChain.
- Built a production-grade RAG pipeline with vector stores and semantic search to ground LLM outputs in enterprise knowledge, improving factual accuracy and traceability of responses.
- Deployed a hybrid LLM architecture combining self-hosted local models (Llama, DeepSeek) for sensitive data processing with Azure OpenAI/OpenAI APIs for high-quality generation where allowed.
- Implemented guardrails, audit logging, and policy-enforcement layers, including full decision traceability so auditors can trace every finding back to its source. Live at yanipro.ai.

ActPilot - AI Browser Agent | *Chrome Extension, Azure OpenAI, Groq* Aug 2026

- Built a Chrome/Edge browser extension (Manifest V3) that reads a page's visible text and a screenshot, then executes real actions on the page (fill fields, click, open links) based on natural-language requests.
- Designed the backend to return one structured JSON response per request, restricting execution to a closed set of actions referencing only elements the model was actually given, preventing hallucinated targets.
- Implemented a hard safety rule so the agent can fill out forms but never submits, pays, deletes, or sends anything unless explicitly instructed.
- Ran on Azure OpenAI's vision-capable model with automatic fallback to Groq on rate-limit errors, keeping the assistant responsive under load.
- Added chart generation and branded PDF export of conversations, with rate limiting and a hosted privacy policy; published to the Microsoft Edge Add-ons store.

Private Data Assistant | *Text-to-SQL, Multi-DB, Read-only Guard* Apr 2026

- Built a text-to-SQL/NoSQL assistant that connects to a user's own live database (PostgreSQL, MySQL, SQL Server, SQLite, or MongoDB) and answers natural-language questions with auto-generated queries.
- Implemented schema-aware RAG to retrieve the relevant tables for each question, with few-shot examples seeded from foreign-key relationships and successful past queries.
- Designed a five-layer, per-engine read-only execution guard (keyword blocking, forced rollback, statement timeouts, row caps) to guarantee the assistant can only ever read data.
- Encrypted stored database credentials with Fernet symmetric encryption, so registered connections are never stored or used in plaintext.
- Added optional chart generation for query results, rendered through a React/Chart.js frontend.

Private Document Assistant | *Agentic RAG, Azure OpenAI, SSE* Jan 2026

- Built a private RAG chatbot for a user's own documents (PDF/DOCX/scanned images with OCR fallback), with strict per-user and per-chat data isolation enforced at the vector-store layer.
- Implemented an agentic tool-calling loop where the model decides per turn whether to retrieve from documents, avoiding spurious grounded-refusals on plain conversational messages.

- Streamed responses token-by-token via Server-Sent Events for a responsive chat experience.
- Used Azure OpenAI for embeddings with a configurable Groq/Azure chat backend, and JWT-based authentication for account and session security.

Private Voice Assistant | *Twilio, Whisper STT, Orpheus TTS* Aug 2025

- Built a runtime-configurable AI voice helpline usable by any business, reachable via real phone calls (Twilio) or a browser hold-to-talk demo, sharing one Groq-powered pipeline.
- Tuned the full conversational turn (speech-to-text, LLM, text-to-speech) down to a 2-2.7 second latency through connection warm-up and cached greetings.
- Implemented automatic caller language detection from speech, adapting the assistant's spoken replies accordingly.
- Designed the Twilio integration around HTTP-based polling rather than WebSocket media streams specifically so the service could run on serverless infrastructure.

WellB - AI-Powered Hotel Management System | *RAG, Llama, OpenAI* Feb. 2025

- Developed comprehensive AI chatbot "WellB" using RAG pipeline with local deployment of Llama and DeepSeek models, integrated with OpenAI GPT for enhanced conversational capabilities.
- Implemented intelligent hotel booking system with multi-model AI approach, enabling natural language booking management through locally deployed and cloud-based LLMs.
- Built emergency alert system with real-time notifications and intelligent priority classification using hybrid AI model architecture.
- Created AI-driven contact management with automated data extraction and contact addition using local Llama model deployment for data privacy.
- Developed booking edit functionality with conflict detection and automated rescheduling using DeepSeek model for complex reasoning tasks.
- Integrated multiple AI services creating cohesive WellB platform with local and cloud-based model orchestration for optimal performance and cost efficiency.

IntelliReport - AI-Powered Report Generation | *Python, NLP, Transformers, FastAPI* Sep. 2024

- Developed comprehensive AI-powered report generation system with automated text correction, topic extraction, and intelligent summarization capabilities using transformer-based models and BERT architectures.
- Implemented advanced topic extraction engine utilizing BERT and custom NLP models to automatically identify and categorize key themes, subjects, and concepts from unstructured text data with hierarchical clustering.
- Built intelligent summary generation system using extractive and abstractive summarization techniques to create concise, coherent summaries while preserving critical information and maintaining contextual coherence.
- Created topic-wise summary functionality that segments documents by identified topics and generates targeted summaries for each category, enabling focused analysis and better content organization with semantic similarity scoring.
- Deployed multi-format report generation with markdown and HTML rendering for direct frontend display and PDF download functionality for seamless user experience across different platforms and devices.
- Built scalable API architecture supporting batch processing with real-time progress tracking and efficient memory management for large document handling and concurrent user requests with load balancing capabilities.

Smart Media Monitoring Application | *Python, PyTorch* August 2023

- Led a comprehensive media monitoring project, involving the development and deployment of multiple AI models, including Speech to Text conversion, sentiment analysis, summary generation, topic modeling, facial recognition, and OCR.
- Worked with both Urdu and English datasets to ensure language versatility and provide valuable insights for the application's users.
- Developed comprehensive text detection and OCR pipeline supporting multi-format document processing including PDFs, images, and tables with multi-language capabilities for Urdu, English, and Arabic text extraction.
- Pioneered facial recognition solutions to identify and track individuals in media content, providing valuable insights for security and analytics.
- Integrated a YouTube Vlogger Scrapper, automating data extraction for content analysis and audience insights.

Smart Surveillance System | *Python, PyTorch and TensorFlow* December 2022

- Led the Smart Surveillance project focused on live camera streams and IP cameras, specializing in AI modules for comprehensive security monitoring.

- Live Object Detection: Implemented real-time object detection models for enhanced security and tracking with high accuracy and low latency performance.
- ANPR (Automatic Number Plate Recognition): Developed a highly accurate ANPR system for vehicle monitoring and identification with multi-language support.
- Geofencing: Created geofencing algorithms to define and enforce geographic boundaries within the camera streams for automated area monitoring.
- Video Summarization: Designed video summarization techniques to condense lengthy footage into concise and informative summaries for efficient review.

AAQAB Project | *Python, Yolo, PyTorch* Feb. 2022

- Led the AAQAB Project, utilizing YOLO (You Only Look Once) and ResNet for object detection in large images with focus on satellite imagery analysis and geospatial intelligence applications.
- Large Image Processing: Worked on processing large images and splitting them based on latitude and longitude coordinates for efficient computational handling and distributed processing across multiple GPU clusters.
- Object Detection with SAHI: Implemented object detection algorithms with a focus on accurate object counting in high-resolution satellite imagery using sliding window approach for enhanced detection accuracy.
- PRSS Image Pipeline: Developed a comprehensive pipeline for PRSS (Pakistan Remote Sensing Satellite) image analysis with automated processing workflows and quality assurance mechanisms for consistent results.

Publications

Meter Digit Recognition Via Faster R-CNN - Muhammad Waqar, M. Waris, et al., M. Yousaf. *International Conference on Robotics and Automation in Industry (ICRAI)*, Oct 2019. DOI: 10.1109/ICRAI47710.2019.8967357.

Technical Skills

Generative AI & Agents: LLM Orchestration, Agentic Tool-Calling, RAG Pipelines, LangChain, LangGraph, Prompt Engineering, OpenAI SDK, Azure OpenAI, GCP/Vertex AI, AWS Bedrock, Groq, Open-Source LLMs (Llama, Mistral, DeepSeek), On-Prem LLM Hosting
ML & Computer Vision: PyTorch, TensorFlow, Transformers, YOLO, scikit-learn, Keras, OpenCV, AWS SageMaker
Voice & Speech AI: Groq Whisper STT, Azure Speech, Text-to-Speech, SeamlessM4T (Meta), Twilio Voice, Real-time Audio Pipelines
Data & Retrieval: PostgreSQL, MySQL, SQL Server, MongoDB, Text-to-SQL, SQLAlchemy, Alembic Migrations
Full-Stack & DevOps: FastAPI, React, TypeScript, Docker, AWS Lambda, JWT Auth, CI/CD (pytest/vitest), Vercel, Git
Leadership & Practice: Team Leadership, System Architecture, Security Engineering, Mentorship, Test-Driven Development